

Computer networking

Assignment - 01

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ICT, CSTU

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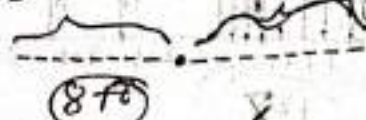
Lecture 9

Computer Networking

IP addressing

Bit $\rightarrow$ binary digit (0, 1)

1 Byte $\rightarrow$ 8 bit

Octet $\rightarrow$ 

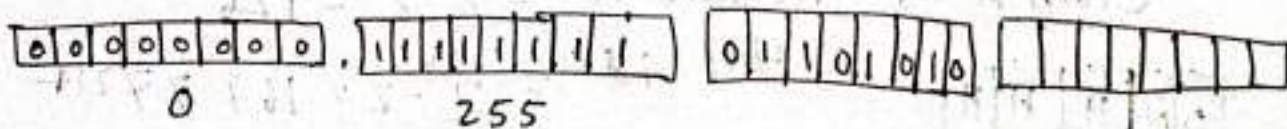
Network
Host

32 bits 4 octets 0-255 5 classes N/H

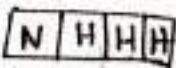
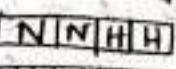
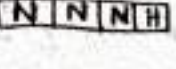
IP-address

IPv4 (4 octets)
 $= 32$ bits

IPv6 (16 octets)
 $= 128$ bits



IP address (class)

class A 
class B 
class C 
class D
class E

① Notation of IP address

② IP range of different class

notation $\left\{ \begin{array}{l} \text{Binary} \rightarrow 10000000 . 00001100 . 00001111 . 00010100 \\ \text{Decimal} \rightarrow 128 . 12 . 15 . 10 \\ \text{Hexadecimal} \rightarrow 80 . C . F . A \end{array} \right.$

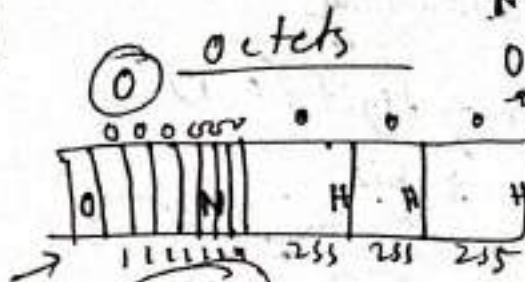
IP Range for different classes:-

128.12.15.10

Class Name

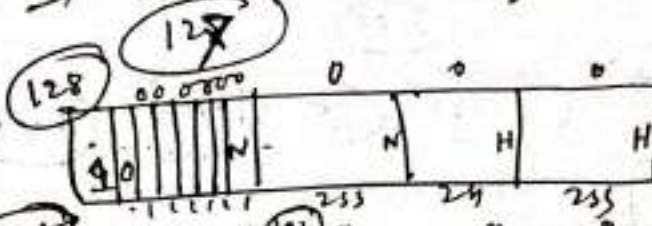
Network Id
0 bits for
to one
Range

class A :



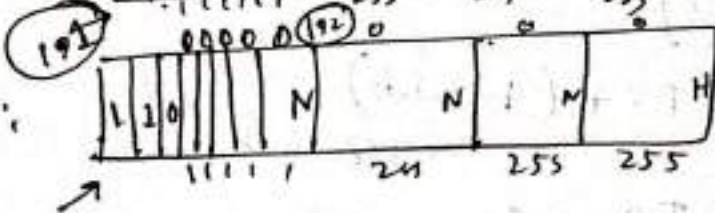
0.0.0.0
↓
~~255.255.255~~
126.255.255.255 ✓

class B :



128.0.0.0
↓
191.255.255.255

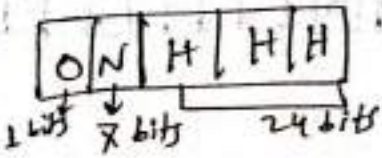
class C :



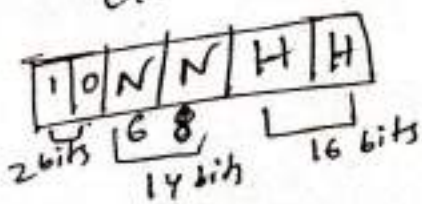
192.0.0.0
↓
223.255.255.255

class A → Network id = $2^7 - 2 = 126$

Host id = $2^{24} - 2 = 16777214$



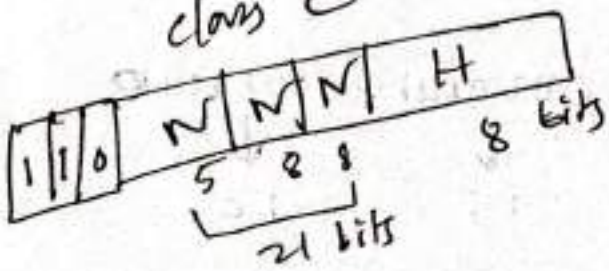
class B



Network id = $2^{14} - 2 = 16384$

Host id = $2^{16} - 2 = 65534$

class C

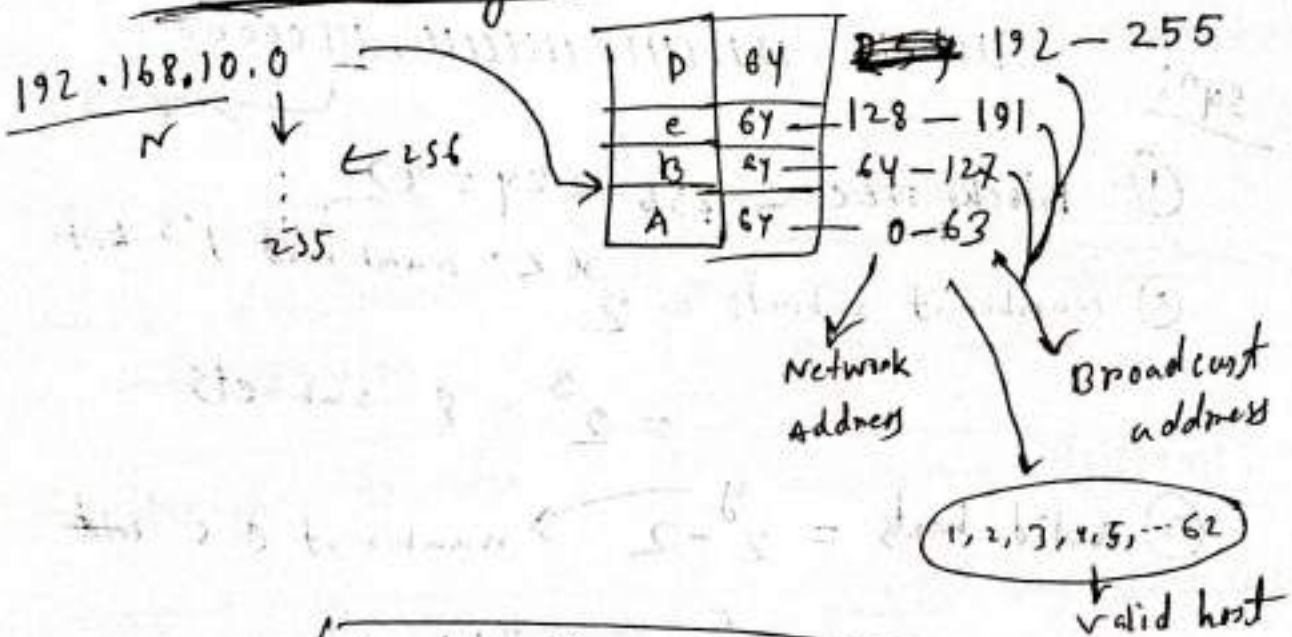


Network id = $2^{21} - 2 = 2097152$

Host id = $2^8 - 2 = 254$

Lecture 4

Subnetting



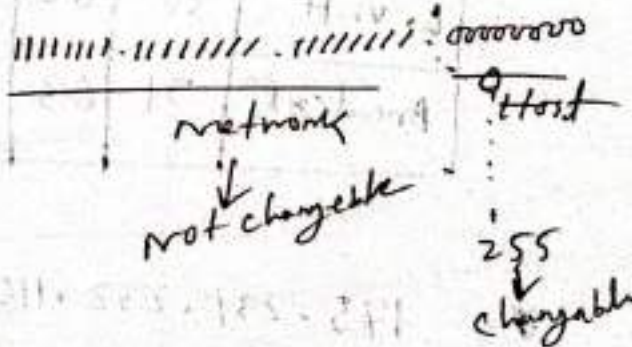
Subnet mask

computers on the network

Host = 0

IP → 192.168.10.0

Subnet mask: 255.255.255.0



Question 1

IP → 192.168.10.0

Subnet mask: 255.255.255.224

Find out

- Block size?
- Number of subnets?
- Number of valid hosts?
- What are the valid subnets on subnet ID?
- What are the first valid host & last valid host?
- What's the Broadcast address?

255 . 255 . 255 . 224

sqn: ||||| ||||| ||||| |||00000
 $\rightarrow x=3$

① Block size = $256 - 224 = 32$

② Number of subnets = $2^x \leftarrow \text{Number of 1's bit}$
 $= 2^3 = 8 \text{ subnets}$

③ valid hosts = $2^y - 2 \rightarrow \text{number of 0's bit}$
 $= 2^5 - 2 = 30$

④ 192.168.10.0 192.168.10.32 $\rightarrow$

Subnet ID	0	32	64	96	128	160	192	224
First V.H	1	33	65	97	129	161	193	225
L. V. H	30	62	94	126	158	190	222	254
Broadcast	31	63	95	127	159	191	223	255

* 175.231.232.116/27 $\leftarrow$ CIDR
 classes Enter domain Routing
 $\rightarrow$ Number of 1's
||||| ||||| ||||| |||00000

Subnetmask: 255.255.255.224

Difference between Connection-oriented and Connectionless service.

Connection-Oriented	Connectionless
connection is established before data transfer	No prior connection is established
Reliable data transfer	Unreliable delivery
Guarantees order of packets	packets may arrive out of order
uses acknowledgements	no acknowledgements
More overhead	Less overhead
Example: TCP	EX: UDP

Differentiate between point to point link and broadcast link:

Aspect	point to point link	Broadcast link
Definition:	A communication link that connects exactly two devices	A communication link where a single sender can transmit to multiple receivers simultaneously.
Number of devices:	Only two devices (one sender one receiver)	More than two devices share the same link
Addressing:	No need for destination addressing	Requires addressing
Bandwidth usage:	Between two devices	Between all devices
Security:	more secure	Less secure
Examples:	Telephone line, leased line one to one	Ethernet LAN, wireless LAN one to many

Differentiate between circuit-switched networks and packet-switched networks.

The primary difference lies in how data is transmitted and how the "path" is established between the sender and receiver.

Feature	Circuit-Switched	Packet-Switched
Connection	Dedicated path established for the duration of the session.	No dedicated path; data is divided into packets.
Path	All data follows the same physical path.	Packets may take different routes to the destination.
Efficiency	Low	High
Example	Traditional Landline Telephone	The Internet (TCP/IP)

Write the subnet address, broadcast address, and valid host range for the following:

IP address: 172.20.75.130

Subnet mask: 255.255.248.0

Soln:

Subnet mask in binary: 11111111.11111111.11110000.00000000

$$\text{Block size} = 256 - 248 = 8$$

Finding the subnet: In the third octet 75 the multiples of 8 are 0, 8, ..., 64, 72, 80. Since 75 falls between 72 and 80, the subnet starts at 72.

Results:

Subnet address: 172.20.72.0

Broadcast address: 172.20.79.255

Valid host range: 172.20.72.1 to

172.20.79.254

An organization is assigned the network address 180.25.40.0. The organization requires six subnets with equal numbers of hosts. Design an appropriate subnetting scheme.

Soln: Given Network 180.25.40.0
Requirement: 6 subnets.

Determine Borrowed Bits: we need $2^n \geq 6$

$$n=3 \quad (2^3=8)$$

default mask for class B $\rightarrow$ 255.255.0.0/16

new subnet mask $\rightarrow$ 255.255.0.0/19

$$\begin{array}{c} \text{|||||} \cdot \text{|||||} \cdot \text{|||||} \cdot \text{|||||} \\ = 255.255.224.0 \end{array}$$

$$\text{Block size} = 256 - 224 = 32$$

<u>Subnet No.</u>	<u>Subnet address</u>	<u>Host Range</u>
1	180.25.0.0	180.25.0.1 - 180.25.31.254
2	180.25.32.0	180.25.32.1 - 180.25.63.254
3	180.25.64.0	180.25.64.1 - 180.25.95.254
4	180.25.96.0	180.25.96.1 - 180.25.127.254
5	180.25.128.0	180.25.128.1 - 180.25.159.254
6	180.25.160.0	180.25.160.1 - 180.25.191.254

Lecture-1

Computer Network

A system of interconnected computers and computerized peripherals such as printers is called computer network.

Classification of computer networks

- ① Geographical span → may be spanned across whole city.
- ② Inter-connectivity → Every single device can be connected to every other device.
- ③ Administration → private or public
- ④ Architecture → Types of computer network
 - Peer-to-peer network
 - client/server network

PAN → Personal Area Network

↳ Bluetooth, infra-red enabled devices, wireless printers, keyboards, TV remotes.
Ex: range up to 10 meters.

LAN → Local Area Network

↳ inside a building or operated under single administrative system is generally term as LAN.

Ex: Schools, College, Universities.

LAN can be wired, wireless or both-forms at once.

MAN → Metropolitan Area Network generally expands throughout a city such as cable TV network.

Ex: Token-ring, ATM, FDDI.

WAN → Wide Area Networks

↳ May be managed by multiple administrations.
Ex: ATM, SONET, Frame Relay.

Internetwork

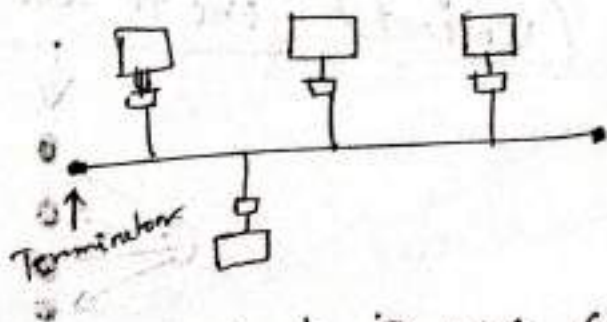
A network of networks is called an internetwork, or simply the internet. uses $\rightarrow$ WWW, FTP, email services, audio and video streaming etc

Internetwork applications

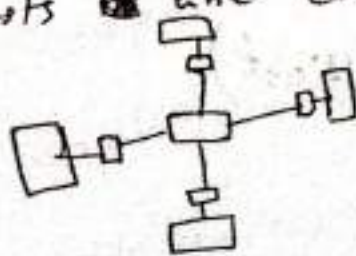
- | | |
|---------------------|------------------------------|
| ① web sites | ⑥ Marketing |
| ② E-mail | ⑦ Networking |
| ③ Instant messaging | ⑧ Resource sharing |
| ④ Blogging | ⑨ Audio and video streaming. |
| ⑤ social media | |

Bus topology

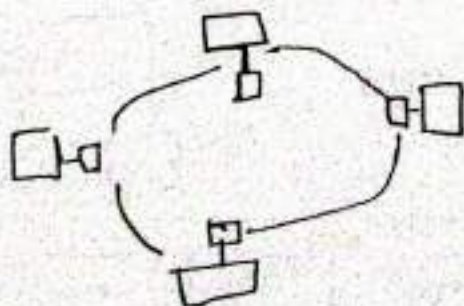
all devices share single communication line or cable.



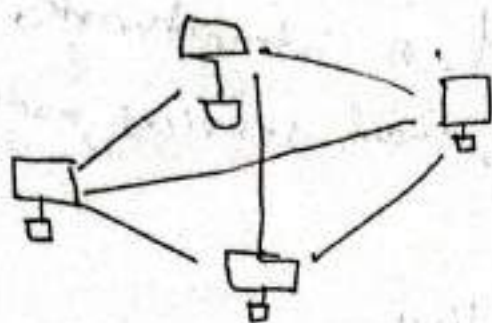
Star topology $\rightarrow$ all hosts are connected to a central device



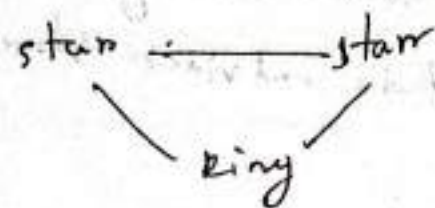
Ring topology $\rightarrow$ each host machine connects to exactly two other machines, creating a circular network structure



Mesh Topology → a host is connected to one or multiple hosts.



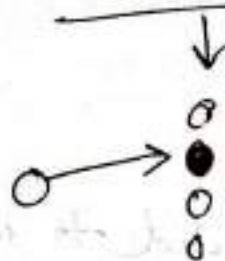
Hybrid Topology → More than One topology



Broadcast link

IP supports the following services:

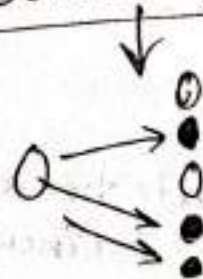
one to one (unicast)



one to all (Broadcast)



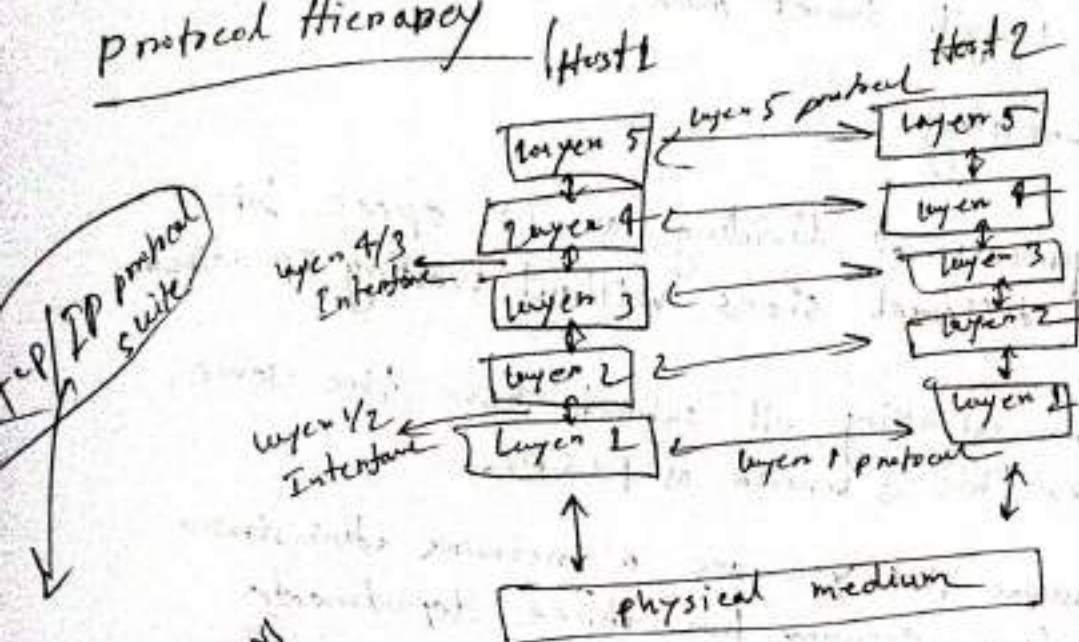
one to several (multicast)



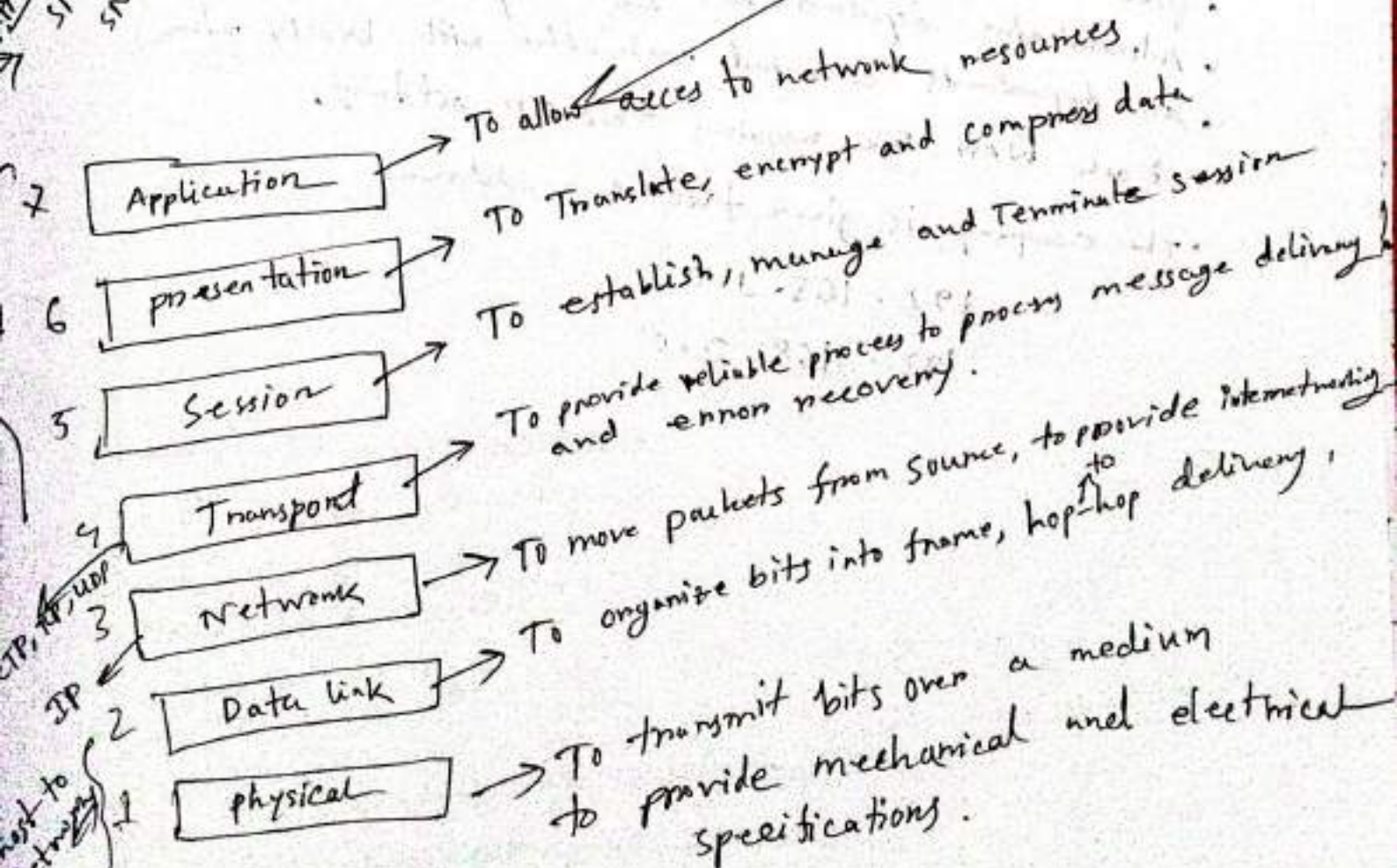
Lecture 2

Transmission Technology → point-to-point link
 → Broadcast link

Protocol Hierarchy



OSI (Open Systems Interconnection) models



VLSM / FLSM

FLSM = Fixed Length Subnet Mask.

VLSM = Variable Length Subnet Mask.

→ classless subnetting

VLSM is a process of dividing an IP space into the subnets of different sizes without wasting IP addresses.

When we perform subnetting, all subnets have the same number of hosts. This is known as FLSM.

Question 230

For example, assume that you are a network administrator of a software firm. Company has three departments connected with WAN links.

- Development department have 24 computers.
- Production department have 52 computers.
- Administrative department have 28 computers.
- All departments connected each other with WAN network.
- Each WAN link requires two IP addresses.
- The company is given two C class addresses.

192.168.1.0

192.168.2.0

Soln:

longest to smallest:

Subnet	segment	Hosts
1	development	74
2	production	52
3	Administration	28
4	WAN link 1	2
5	WAN 2	2
6	WAN link 3	2

For subnet 1

74 Hosts

$$2^7 = 128$$

network host

$$32 - 7 = 25$$

$$CIDR = /25$$

$$\text{network ID} = 192.168.1.0$$

$$\text{First host ID} = 192.168.1.1$$

$$\text{Last Host ID} = 192.168.1.126$$

$$\text{Broadcast ID} = 192.168.1.127$$

$$\text{subnet mask} = 255.255.255.128$$

For subnet 2

52 Hosts

$$2^6 = 64$$

$$CIDR = /26$$

subnet mask:

$$255.255.255.192$$

$$\text{network IP} = 192.168.1.128$$

$$\text{FHI} = 192.168.1.129$$

$$\text{LHI} = 192.168.1.190$$

$$\text{Broadcast ID} = 192.168.1.191$$

for subnet 3

28 Hosts

$$2^5 = 32$$

$$CIDR = /27$$

$$\text{Subnet mask} = 255.255.255.224$$

$$\text{network ID} = 192.168.1.192$$

$$\text{Broadcast} = 192.168.1.223$$

WAN-1
segment $n=2 \rightarrow CIDR = /30$

Subnet mask = 255.255.255.252

Network ID = 192.168.1.224

Broadcast = 192.168.1.227

WAN-2, WAN-3
Similarly $\rightarrow \checkmark$

Broadcast ID = 192.168.1.235

We have assigned IP addresses to all segments;
still we have 20 addresses available. This is the
magic of VLSM.

Lecture 8

Differentiate between Virtual circuit and datagram.

<u>Feature</u>	<u>VC</u>	<u>Datagram</u>
Connection Type	Connection-Oriented	Connectionless
Path selection	All packets follow a fixed, pre-defined path for the entire session	Each packet is treated independently and can take different path
Flexibility	less flexibility	More flexibility
Packet sequencing	arrive in the connect order	packets may arrive out of order
Header information	contains a short Virtual Circuit Identifier (VCI)	contains full source and destination IP address in every packet.
Resource Reservation	Reserved	no reserved
Failure handling	If a node/link on the path fails, the entire connection is lost.	If a path fails, packets are simply rerouted through a different path.
Example	ATM (Asynchronous Transfer Mode).	IP (Internet protocol) UDP

Intra-Inter Domain Routing

⇒ Routing inside an autonomous system is called intra-domain routing.
 Routing between autonomous systems is called inter-domain routing.

Nonadaptive = static routing because it does not respond to failures

Adaptive algorithms = In contrast, change their routing decisions and traffic as well.

Shortest path algorithm

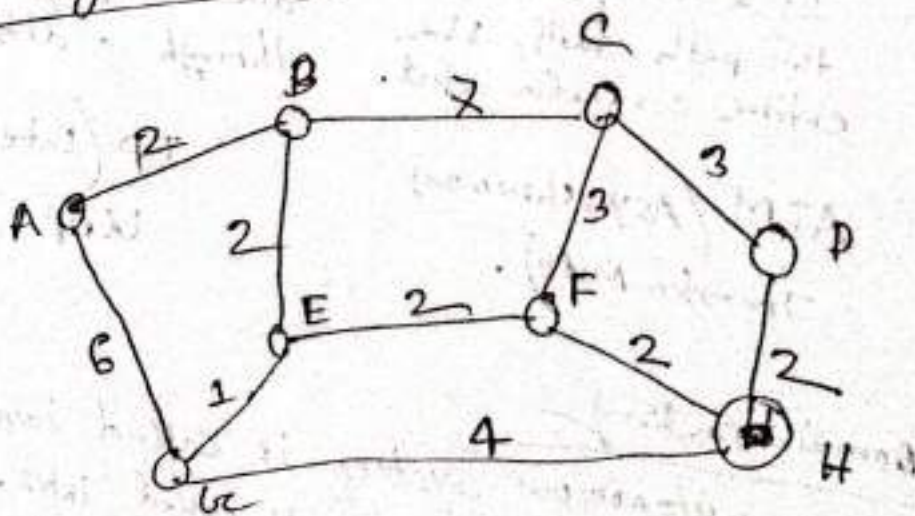
Flooding is a simple computer network routing algorithm in which every incoming packet is sent through every outgoing link except the one it arrived on.

⇒ Inefficient because one node may receive multiple copies of message.
waste no. of duplicate copy made.

Three technique

- ✓ ① Hop counter
- ✓ ② Sequence number
- ③ Selection method.

Dijkstra Algorithm



Difference between flooding and broadcasting

Feature	Broadcasting	Flooding
Scope	Limited to a specific network	can spread across the entire network
duplicate packets	A node receives only one copy of the packet	A node may receive multiple copies of the same packet from different neighbours.
Efficiency	More efficient	inefficient
Mechanism	uses a specific broadcast address	uses a standard outgoing interface logic
usage	used for address resolution or DHCP	used in routing protocols, (OSPF)

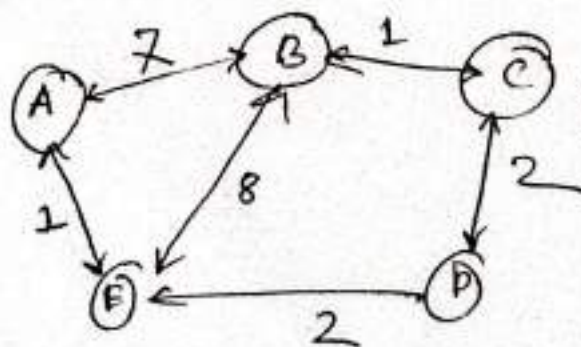
* What is 'routing loop' and 'convergence'?

⇒ Routing loop occurs when a data packet is routed through the same route again and again in an endless circle.

⇒ Network is considered as 'converged' once all the routers on a network are updated with the changes.

Distance vector routing

→ slow convergence



step 1 Initial state

Initial node	distance to node				
	A	B	C	D	E
A	0	7	∞	∞	1
B	7	0	1	∞	8
C	∞	1	0	2	∞
D	∞	∞	2	0	2
E	1	8	∞	2	0

step 1 neighbours

step 2

A

Dest	dist	next hop
A	0	A
B	6	A, D, C
C	5	E, D
D	3	E
E	1	A

B

Dest	dist	next
A	6	C, D, E
B	0	B
C	1	C
D	3	C
E	5	C, D

C

Dest	dist	next
A	5	D, E
B	2	B
C	0	C
D	2	D
E	4	D

D

Dest	dist	next
A	3	E
B	3	C
C	2	C
D	0	D
E	2	E

E

Dest	dist	next
A	1	A
B	5	D, C
C	4	D
D	2	D
E	0	E

final state

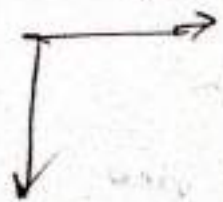
Info & dist	Distance to node				
	A	B	C	D	E
A	0	6	5	3	1
B	6	0	1	3	5
C	5	1	0	2	4
D	3	3	2	0	2
E	1	5	4	2	0

node B's distance vector

(B)

Dest	next hop		
	A	E	C
A	7	9	6
C	12	12	1
D	10	10	3
E	8	8	5

Handling failure



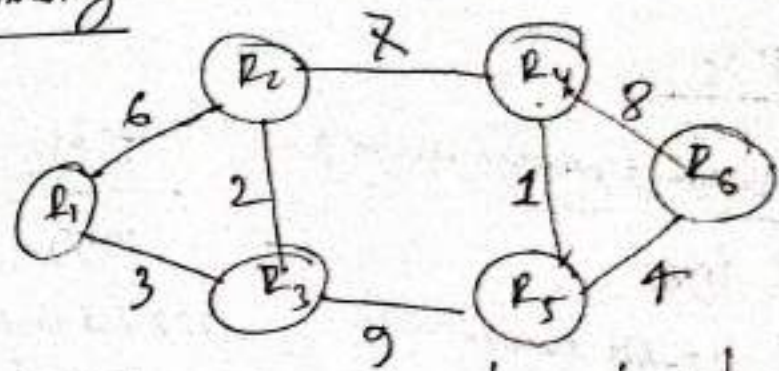
Link state routing Algorithm How to work:

- ① Link state - router knows entire network topology and ~~computes~~ computes shortest path routing.
- ② Each node send its neighbor information known as link state advertisement.
- ③ LS advertisement sent on periodic basis.
- ④ Once node has complete view of network then it populate its routing table by using Dijkstra algorithm for shortest path.

Link state routing Algo. VS Distance vector algo.

<u>Feature</u>	<u>Distance Vector</u>	<u>LS routing</u>
<u>Algorithm</u>	Bellman-Ford	Dijkstra
Topology view	only knows neighbors (partial)	knows entire network (full map)
updates	periodic (Even if no change)	Event-driven (only when a link changes)
Convergence	slow	Fast.
CPU/Memory	Low requirement	High requirement
Traffic	High	Low

Link state routing



Flooding

R1	R2	R3	R4	R5	R6
seq no					
TTL					
R2 6	R1 6	R3 2		R3 9	R4 1
R3 3	R4 7			R5 9	R6 9

By using Dijkstra

	R1	R2	R3	R4	R5	R6
R1		6	(3)	∞	∞	∞
R2	6			∞	12	∞
R3	(5)	3		∞	12	∞
R4	∞	∞	∞		12	∞
R5	∞	12	12	12		21
R6	∞	∞	∞	∞	21	



Dest	Dist	Next Hop
R1	0	R1
R2	5	R3
R3	3	R1
R4	12	R3 R2
R5	12	R3
R6	16	R3 R5

IPv4 VS IPv6

Feature	Internet protocol version 4	IPv6
Deployed	1981	1999
Address size	32-bit number	128-bit number
Address Format	Dotted Decimal notation 192.149.252.76	Hexadecimal notation 3FFF:F100:0234:ABCD: 0123:4567:8901:ABCD
prefix notation	192.149.0.0/24	3FFE:F200:0234::/48
Number of Address	$2^{32} = 4 \text{ billion addresses}$	$2^{128} = (2^{32})^4 = 256 \text{ billion}$

• IPv6 defines three types of address → unicast, anycast, multicast

CIDR-Route Aggregation (Classless Inter-Domain)

CIDR Route aggregation is a method used to minimize the number of routing table entries by combining multiple contiguous network addresses into a single, broader address.

✳ Suppose we have four IP subnets on the four LAN interfaces of our router.

- 210.6.0.0/24
- 210.6.1.0/24
- 210.6.3.0/24
- 210.6.3.0/24

Summarize these networks into a single route that can be advertised across the WAN / calculate the supernet address for the given networks / Summarize them into a

Sol:

The four subnets in binary are...

11010010 . 00000110 . 00000000 . 00000000
 11010010 . 00000110 . 00000001 . 00000000
 11010010 . 00000110 . 00000010 . 00000000
 11010010 . 00000110 . 00000011 . 00000000

The common digits starting from the left are...

11010010 . 00000110 . 00000000

since there are 22 common digits then the single route is

210.6.0.0/22

Ans

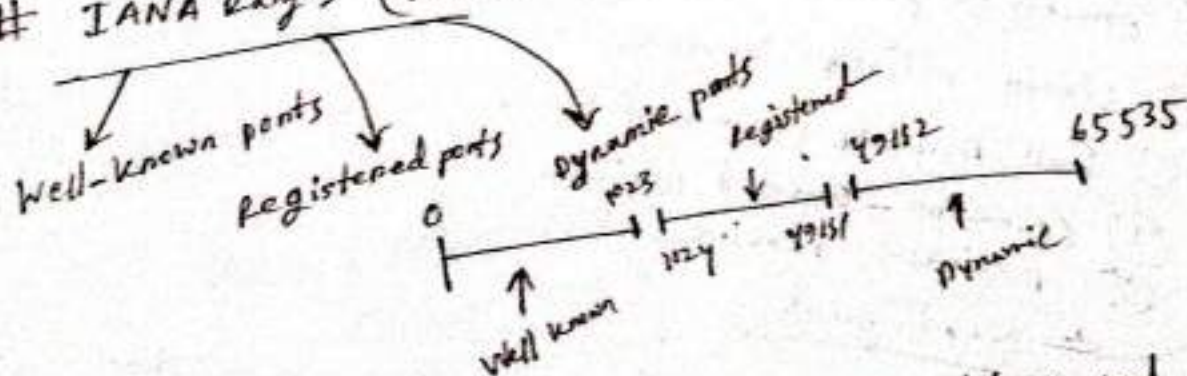
All math are same of this type

lecture-11

Client-server paradigm

* A process on the local host, called a client, needs services from a process usually on the remote host, called a server.

IANA Ranges (Internet Assigned Number Authority)

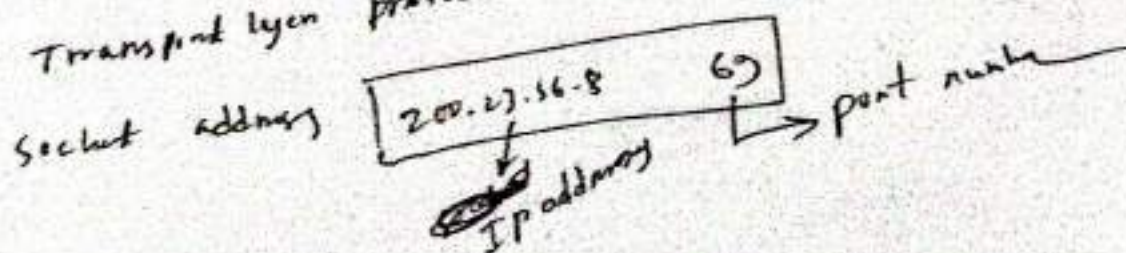


Socket address → The combination of an IP address and a port number is called a socket address.

* process to process delivery needs two identification, IP address and the port number

at each end to make a connection.

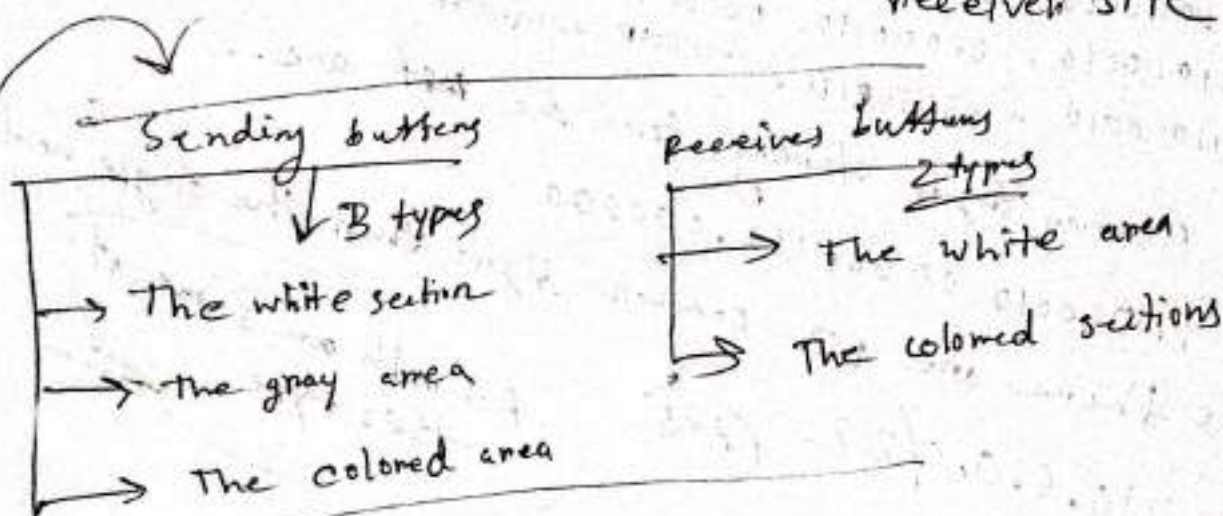
Transport layer protocol needs a pair of socket address → client, server



~~Mux-Dem~~ Multiplexing and demultiplexing

sender site

receiver site



TCP and UDP protocols establishing a TCP connection.

* TCP services ^{mainly by} Transport layer

Services offered by TCP at the application layer:

- ① process to process communication
- ② stream delivery service
- ③ Full Duplex communication
- ④ Connection-oriented service
- ⑤ Reliable service

next point

TCP connection-oriented transmission requires 3 phases:

- ① Connection establishment
- ② Data transfer
- ③ Connection Termination

Three-way handshake protocol is the fundamental process used in networking to establish a reliable, connection-oriented communication session between a client and a server before any actual data is exchanged.

⇒ Works in 3 way

① SYN (synchronize) / FINN segment
 the client starts the process by sending a SYN packet to the server.

• purpose : start a connection (X)

• state : The client moves into the SYN-SENT state.

FINN + Ack segment

② SYN-ACK (Synchronize - Acknowledgement) → server receives the request

• Purpose : received request
 $ACK = X + 1$ and new sequence number Y

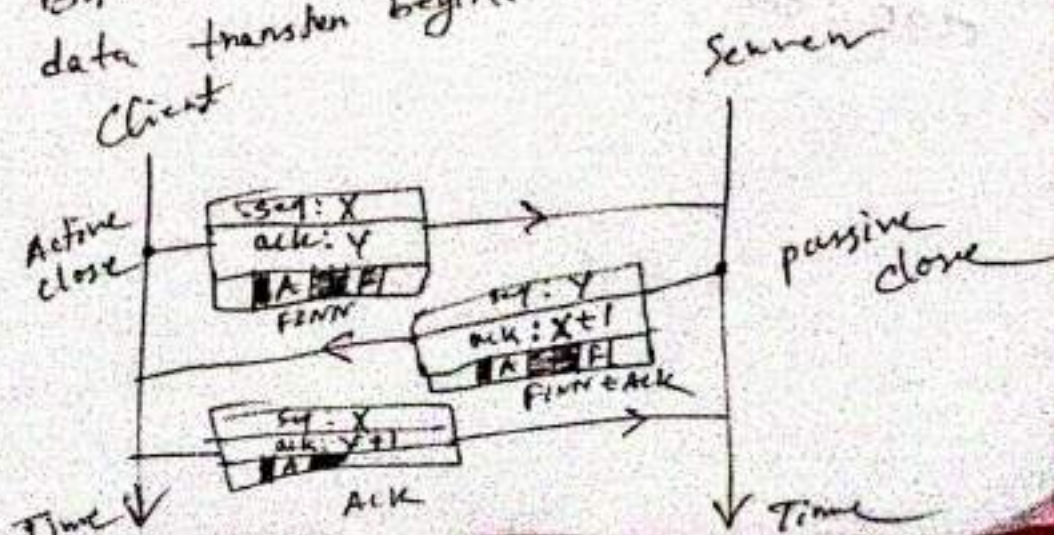
• state : The server moves into the SYN-RECEIVED state.

③ ACK (Acknowledge)

Client receives the server's response and sends a final ACK packet.

• Purpose : $ACK = Y + 1$; received sequence number.

• State : Both devices move into the ESTABLISHED state, and data transfer begins.



Short to long IPv6

An IPv6 address always consists of 8 groups of 4 hexadecimal digits each (128 bits total)

The two steps to expand

- ① Fill in the missing groups (::)
- ② Restore leading zeros

Examples Short
::1 (loopback)

:: (unspecified)

2001:db8:0:1::5

fe80::215:5dff

Long
0000:0000:0000:0000:0000:0000:0000:0001

0000:0000:0000:0000:
0000:0000:0000:0000

2001:0db8:0000:0001:
0000:0000:0000:0005

fe80:0000:0000:0000:
0000:0000:0215:5dff

long to short IPv6

Long

CEDF:BF76:0000:0000:
009E:0000:3025:DF12

Short

CEDF:BF76::009F:0:
3025:DF12

Error detection and connection in TCP is achieved through the use of three simple tools: Checksum, acknowledgment, Time-out, retransmission.

⇒ Checksum → is used to check for a corrupted segment

① If the segment is corrupted, it is discarded by TCP and considered as lost.

② TCP uses 16 bit Checksum that is mandatory in every segment.

⇒ Acknowledgement

TCP uses acknowledgments to confirm the receipt of data segments.

Ack segments do not consume sequence numbers and are never acknowledged. Control segments don't carry data but consume sequence numbers and are also acknowledged.

⇒ Retransmission

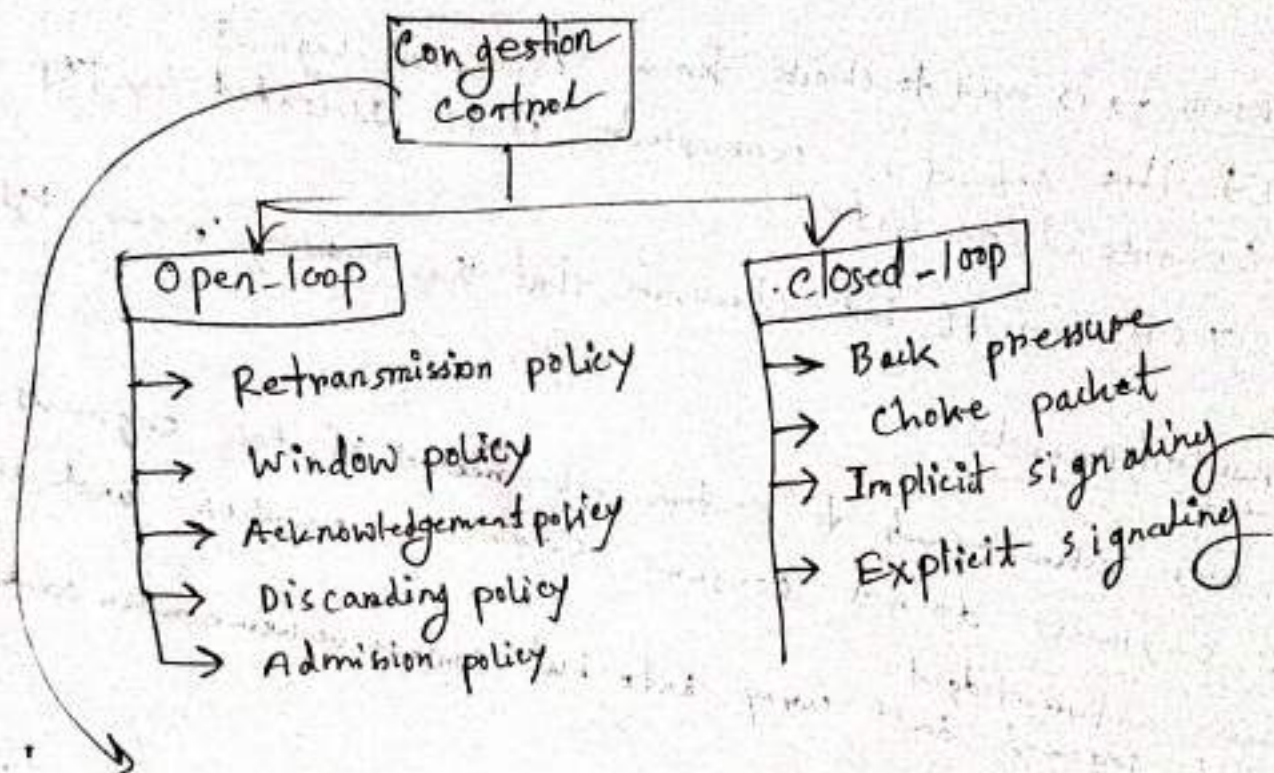
The heart of the error control mechanism is the retransmission of segments.

A segment is retransmitted on two occasions:

- ① When a retransmission timer expires on
- ② When the sender receives three duplicate ACKs.

* Lecture 13

Congestion control categories



Main Focus

- ① we try to avoid traffic congestion
- ② try to create an appropriate environment for the traffic.

We need aware of two issues:

① if the rate of packet arrival is higher than the packet processing rate, the input queues become longer and longer.

② less the ^{output} queues become longer and longer.

Lecture 14

ALOHA

① Each network range $\Rightarrow 0$ to $2K-1$
a multiple

② is randomly chosen and multiplied by TP
 $TP = \text{maximum propagation Time}$

Example 1 The stations on a wireless ALOHA network are a maximum of 600 km apart. If we assume that signals propagate at 3×10^8 m/s, we find

$$T_P = (600 \times 10^3) / (3 \times 10^8) \\ = 2 \text{ ms}$$

Now we can find the value of T_B for different values of K .

Soln For $K=1$, the range is $\{0, 1\}$. This means that T_B can be ~~0 or 6~~ ms, based on the outcome of the random variable.

For $K=2$, the range is $\{0, 1, 2, 3\}$. This means that T_B can be 0, 2, 4, or 6 ms,

For $K=3$, the range is $\{0, 1, 2, 3, 4, 5, 6, 7\}$. This means that T_B can be 0, 2, 4, 6, 8, 10, 12, or 14 ms, based on the outcome of the random variable.

Ex-2 A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the requirement to make this frame collision-free?

Soln: Average frame transmission time T_{fr} is $200 \text{ bits} / 200 \text{ kbps}$ or 1 ms. The vulnerable time is $2 \times 1 \text{ ms} = 2 \text{ ms}$

ALOHA

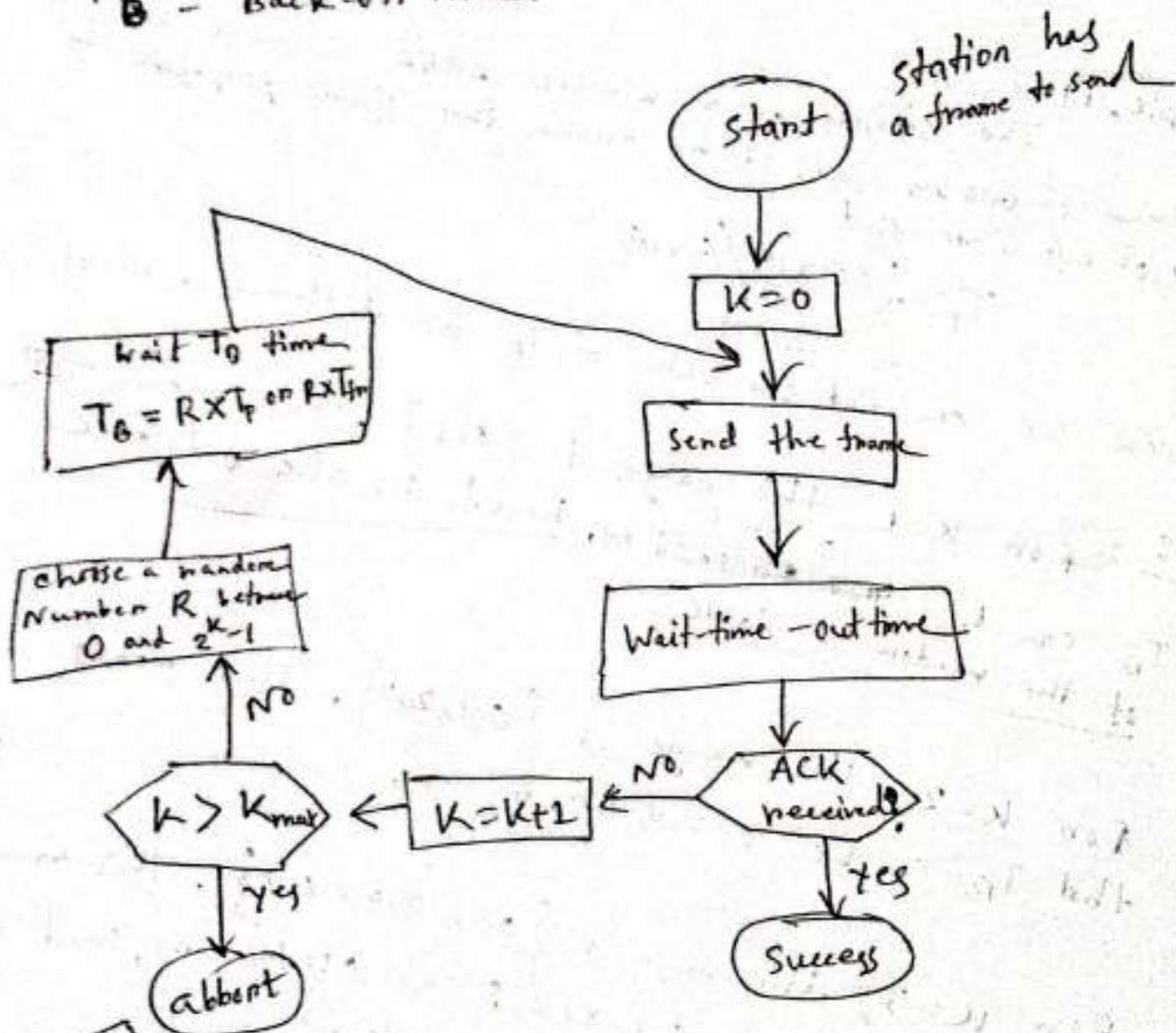
Sublayer $\rightarrow$ ALOHA
 $\rightarrow$ Slotted ALOHA

K = Number of Attempts

T_p $\rightarrow$ Maximum propagation time

T_{fr} = Average transmission time for a frame

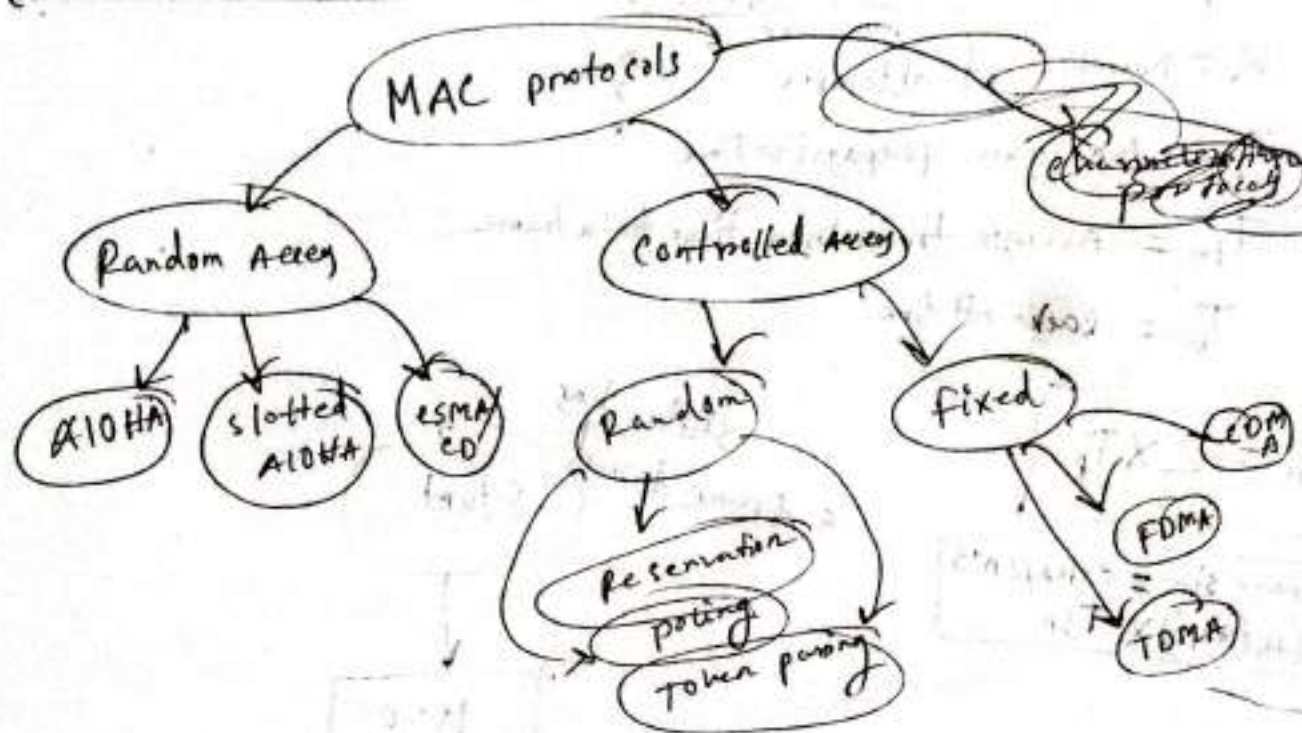
T_B = Back-off time



K_{max} is normally 15

* Slotted ALOHA was invented to improve the efficiency of pure ALOHA.
Slotted ALOHA vulnerable time = T_{fr}

Taxonomy of Multiple access protocols (MAC protocols)



Lecture 16

A network using CSMA/CD has a bandwidth of 10 Mbps.
 If the maximum propagation time is 25.6 μ sec,
 what is the minimum size of the frame?

Soln: The frame transmission time is $T_{fr} = 2 \times T_p = 51.2 \mu$ sec
 This means, in the worst case, a station needs to transmit
 for a period of 51.2 μ s to detect the collision.
 The minimum size of the frame is $10 \text{ Mbps} \times 51.2 \mu$ s
 $= 512 \text{ bits or } 64 \text{ bytes}$

This is actually the minimum size of the frame for
 standard Ethernet.

CSMA/CD Flow Diagram:

K = Number of attempts

T_p = Maximum propagation Time

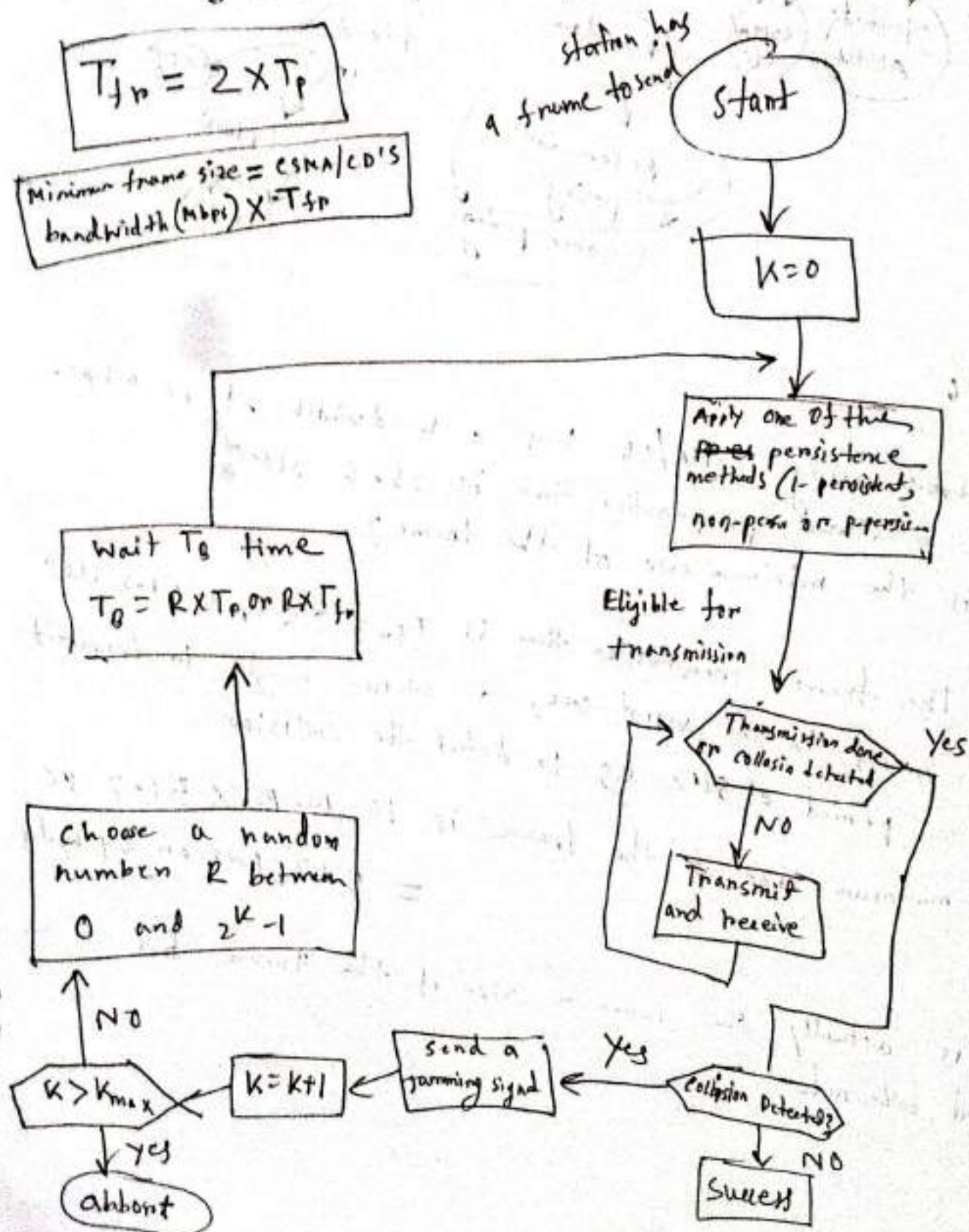
T_{fr} = Average transmission time for a frame

T_B = Back-off time

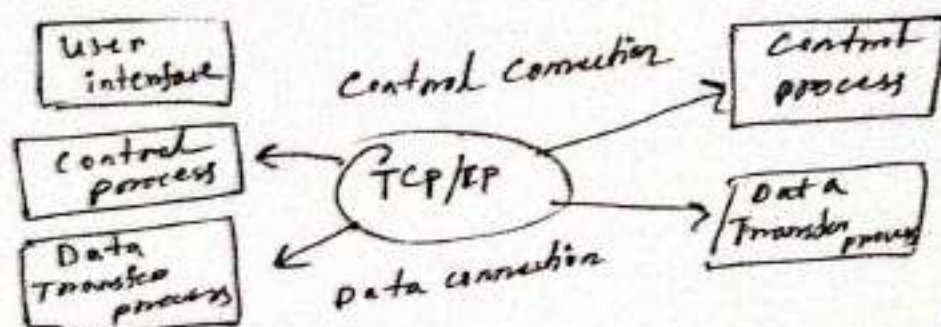
$$T_{fr} = 2 \times T_p$$

Minimum frame size = CSMA/CD's
bandwidth (Mbps) $\times T_{fr}$

Station has
a frame to send



Basic model of FTP



Client

User agent (ua)

Mail transfer agent (MTA)

SMTP:

allows more complex system
the TCP/IP protocol that supports electronic mail on the internet
is called SMTP.

SMTP supports

- ① sending single message to one or more recipients
- ② sending messages include text, audio, video
- ③ sending message to users on networks outside the internet

MIME

allows non ASCII data to ASCII data and vice-versa
not a mail protocol and can not replace SMTP.
it is only an extension to SMTP.